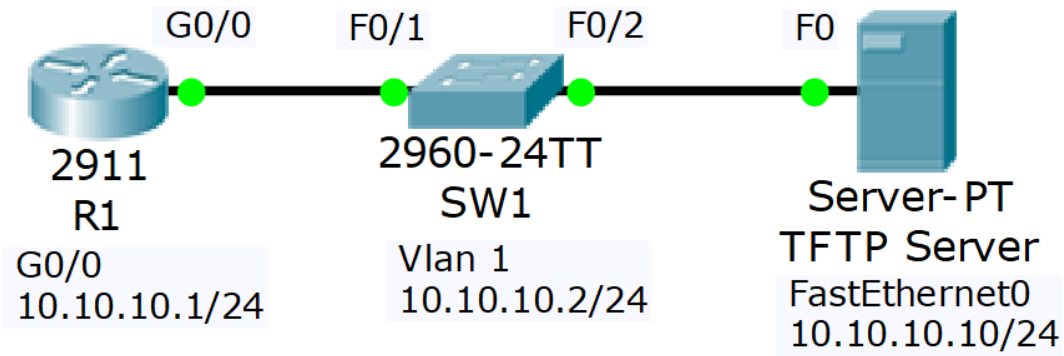
15 Cisco Device Management - Lab Exercise

In this lab you will perform a factory reset, password recovery, configuration backup, and system image backup and recovery on a Cisco router. You will also perform an IOS upgrade on a Cisco switch.

Use Cisco Packet Tracer for this exercise. The generic server in Packet Tracer (as shown in the topology diagram below) has built-in TFTP server software.

Lab Topology



Load the Startup Configurations

Download the '15 Cisco Device Management.zip' file here. *Extract* the project .pkt file then open it in Packet Tracer. Do not try to open the project from directly inside the zip file.

Factory Reset

1) View the running configuration on R1. Note that the hostname and interface have been configured. Copy and paste the configuration into Notepad on your PC.

2) Factory reset R1 and reboot

3) Watch the boot up process as the router boots

4) The router should boot into the Setup Wizard. Exit out of the wizard and then confirm the startup and running configurations are empty.

5) Paste the configuration you copied into Notepad back into the R1 router configuration and save.

Password Recovery

6) Set the enable secret ‘Flackbox1’ on R1 and save the running-configuration.

7) Configure the router to boot into the rommon prompt on next reload with an appropriate command and reboot the router.

8) In rommon mode, configure the router to ignore the startup-config when booting up, and reload the router.

9) The router should boot into the Setup Wizard. Exit out of the wizard.

10) What do you expect to see if you view the running and startup configurations? Confirm this.

11) Copy the startup config to the running config. Do not miss this step or you will factory reset the router!

12) Verify the status of interface GigabitEthernet0/0. Why is it down?

13) Bring interface GigabitEthernet0/0 up.

14) Remove the enable secret.

15) Ensure the router will reboot normally on the next reload and that you will be able to access the router.

16) Reboot the router and confirm it has the expected configuration.

Configuration Backup

17) Backup the running configuration to Flash on R1. Use a suitable name for the backup file. Verify the configuration has been backed up.

18) Backup the R1 startup configuration to the TFTP server. Use a suitable name for the backup file. Verify the configuration has been backed up.

IOS System Image Backup and Recovery

19) Backup the IOS system image on R1 to the TFTP server. Verify the configuration has been backed up.

20) Delete the system image from Flash and reload.

21) Use Internet search to find system recovery instructions for your model of router. Recover the system image using the TFTP server.

IOS Image Upgrade

22) Verify SW1 is running C2960 Software (C2960-LANBASE-M), Version 12.2(25)FX

23) Use the TFTP server to upgrade to c2960-lanbasek9-mz.150-2.SE4.bin

24) Reboot and verify the switch is running the new software version.

15 Cisco Device Management - Answer Key

In this lab you will perform a factory reset, password recovery, configuration backup, and system image backup and recovery on a Cisco router. You will also perform an IOS upgrade on a Cisco switch.

Factory Reset

1) View the running configuration on R1. Note that the hostname and interface have been configured. Copy and paste the configuration into Notepad on your PC.

```
R1#sh run
Building configuration...

Current configuration : 696 bytes
!
hostname R1
!
interface GigabitEthernet0/0
ip address 10.10.10.1 255.255.255.0
duplex auto
speed auto
```

2) Factory reset R1 and reboot

```
R1#write erase
Erasing the nvram filesystem will remove all configuration
files! Continue? [confirm]
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
R1#reload
Proceed with reload? [confirm]
```

3) Watch the boot up process as the router boots

```
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)

Readonly ROMMON initialized

IOS Image Load Test
___________________
```

```
Digitally Signed Release Software

Self decompressing the image :
############################################################
############### [OK]
```

4) The router should boot into the Setup Wizard. Exit out of the wizard and then confirm the startup and running configurations are empty.

```
--- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: no

Router>enable
Router#show run
Building configuration...
hostname Router
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
shutdown

Router#show start
startup-config is not present
```

5) Paste the configuration you copied into Notepad back into the R1 router configuration and save.

```
Router#configure terminal
Router(config)#hostname R1
R1(config)#!
R1(config)#interface GigabitEthernet0/0
R1(config-if)# ip address 10.10.10.1 255.255.255.0
R1(config-if)# duplex auto
R1(config-if)# speed auto
R1(config-if)# no shutdown
R1(config-if)#!
R1(config-if)#line con 0
R1(config-line)# exec-timeout 30 0
R1(config-line)#end
R1#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
```

Password Recovery

6) Set the enable secret ‘Flackbox1’ on R1 and save the running-configuration.

```
R1(config)#enable secret Flackbox1
R1(config)# do copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
R1(config)#
```

7) Configure the router to boot into the rommon prompt on next reload with an appropriate command and reboot the router.

```
R1(config)#config-register 0x2120
R1(config)#end
R1#reload
Proceed with reload? [confirm]
```

8) In rommon mode, configure the router to ignore the startup-config when booting up, and reload the router.

```
rommon 1 > confreg 0x2142
rommon 2 > reset
```

9) The router should boot into the Setup Wizard. Exit out of the wizard.

```
--- System Configuration Dialog ---
Continue with configuration dialog? [yes/no]: no
```

10) What do you expect to see if you view the running and startup configurations? Confirm this.

The running configuration should be empty because the router bypassed loading the startup config on boot up. The startup config should remain unchanged and all previous configuration should still be there.

```
Router#sh run
Building configuration...

hostname Router
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto

Router#sh start
!
hostname R1
!
enable secret 5 $1$mERr$J2XZHMOgpVVXdLjC9lYtE1
!
interface GigabitEthernet0/0
ip address 10.10.10.1 255.255.255.0
duplex auto
speed auto
```

11) Copy the startup config to the running config. Do not miss this step or you will factory reset the router!

```
Router#copy start run
Destination filename [running-config]?
```

12) Verify the status of interface GigabitEthernet0/0. Why is it down?

```
R1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.10.1 YES NVRAM administratively down down
GigabitEthernet0/1 unassigned YES NVRAM administratively down down
GigabitEthernet0/2 unassigned YES NVRAM administratively down down
Vlan1 unassigned YES NVRAM administratively down down

R1#show run
! truncated
interface GigabitEthernet0/0
ip address 10.10.10.1 255.255.255.0
duplex auto
speed auto
shutdown
```

Router interfaces are shut down by default. Because ‘no shutdown’ does not explicitly appear in the startup-configuration for the interface, when the startup-configuration is copied to the running-configuration the default value is applied and the interface is in the shutdown state.

13) Bring interface GigabitEthernet0/0 up.

```
R1(config)#interface g0/0
R1(config-if)#no shutdown
```

14) Remove the enable secret.

```
R1(config)#no enable secret
```

15) Ensure the router will reboot normally on the next reload and that you will be able to access the router.

```
R1(config)#config-register 0x2102
R1(config)#end
R1#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
```

16) Reboot the router and confirm it has the expected configuration.

```
R1#reload
Proceed with reload? [confirm]

R1>en
R1#show run
Building configuration...
hostname R1
!
interface GigabitEthernet0/0
ip address 10.10.10.1 255.255.255.0
duplex auto
speed auto
```

```
R1#sh ip int brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.10.1 YES NVRAM up up
GigabitEthernet0/1 unassigned YES NVRAM administratively down down
GigabitEthernet0/2 unassigned YES NVRAM administratively down down
Vlan1 unassigned YES NVRAM administratively down down
```

Configuration Backup

Important: Filenames are case sensitive – you must enter them *exactly* as shown (**c**2900 is different to **C**2900).

17) Backup the running configuration to Flash on R1. Use a suitable name for the backup file. Verify the configuration has been backed up.

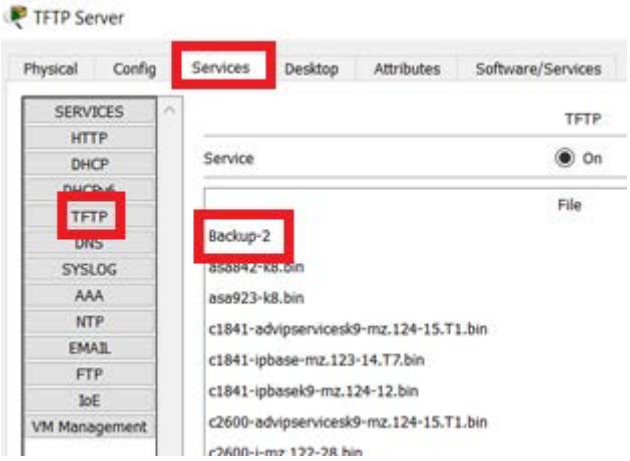
```
R1#copy run flash
Destination filename [running-config]? Backup-1
Building configuration...
[OK]

R1#show flash

System flash directory:
File Length Name/status
5 728 Backup-1
3 33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
2 28282 sigdef-category.xml
1 227537 sigdef-default.xml
[33848315 bytes used, 221895685 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

18) Backup the R1 startup configuration to the TFTP server. Use a suitable name for the backup file. Verify the configuration has been backed up.

```
R1#copy start tftp
Address or name of remote host []? 10.10.10.10
Destination filename [R1-confg]? Backup-2
Writing startup-config....!!
[OK - 698 bytes]
698 bytes copied in 3.007 secs (242 bytes/sec)
```



IOS System Image Backup and Recovery

19) Backup the IOS system image on R1 to the TFTP server. Verify the configuration has been backed up.

```
R1#show flash

System flash directory:
File Length Name/status
5 728 Backup-1
3 33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
2 28282 sigdef-category.xml
1 227537 sigdef-default.xml
[33848315 bytes used, 221895685 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)

R1#copy flash tftp
Source filename []? c2900-universalk9-mz.SPA.151-4.M4.bin
Address or name of remote host []? 10.10.10.10
Destination filename [c2900-universalk9-mz.SPA.151-
4.M4.bin]?

Writing c2900-universalk9-mz.SPA.151-
4.M4.bin...!!!
```

20) Delete the system image from Flash and reload.

```
R1#delete flash:c2900-universalk9-mz.SPA.151-4.M4.bin
Delete filename [c2900-universalk9-mz.SPA.151-4.M4.bin]?
Delete flash:/c2900-universalk9-mz.SPA.151-4.M4.bin?
[confirm]

R1#reload
Proceed with reload? [confirm]
Boot process failed...
The system is unable to boot automatically. The BOOT
environment variable needs to be set to a bootable
image.
rommon 1 >
```

21) Use Internet search to find system recovery instructions for your model of router. Recover the system image using the TFTP server.

http://www.cisco.com/c/en/us/td/docs/routers/access/1900/software/configuration/guide/Software_Configuration/appendixCrommon.html is the first hit when searching for 'Cisco 2900 rommon recovery'.

Go to the "Recovering the System Image (tftpdnld)" section.

The 'tftpdnld' command has built-in help which is displayed when entering the command in rommon mode:

```
rommon 1 > tftpdnld

Missing or illegal ip address for variable IP_ADDRESS
Illegal IP address.

usage: tftpdnld
Use this command for disaster recovery only to recover an image via
TFTP.
Monitor variables are used to set up parameters for the transfer.
(Syntax: "VARIABLE_NAME=value" and use "set" to show current
variables.)
"ctrl-c" or "break" stops the transfer before flash erase begins.

The following variables are REQUIRED to be set for tftpdnld:
IP_ADDRESS: The IP address for this unit
IP_SUBNET_MASK: The subnet mask for this unit
DEFAULT_GATEWAY: The default gateway for this unit
TFTP_SERVER: The IP address of the server to fetch from
TFTP_FILE: The filename to fetch

The following variables are OPTIONAL:
TFTP_VERBOSE: Print setting. 0=quiet, 1=progress(default), 2=verbose
TFTP_RETRY_COUNT: Retry count for ARP and TFTP (default=7)
TFTP_TIMEOUT: Overall timeout of operation in seconds (default=7200)
```

Use **ALL CAPITAL LETTERS** for this configuration:

```
rommon 2 > IP_ADDRESS=10.10.10.1
rommon 3 > IP_SUBNET_MASK=255.255.255.0
rommon 4 > DEFAULT_GATEWAY=10.10.10.1
rommon 5 > TFTP_SERVER=10.10.10.10
rommon 6 > TFTP_FILE=c2900-universalk9-mz.SPA.151-4.M4.bin
rommon 7 > tftpdnld

IP_ADDRESS: 10.10.10.1
IP_SUBNET_MASK: 255.255.255.0
DEFAULT_GATEWAY: 10.10.10.1
TFTP_SERVER: 10.10.10.10
TFTP_FILE: c2900-universalk9-mz.SPA.151-4.M4.bin
Invoke this command for disaster recovery only.
WARNING: all existing data in all partitions on flash will
be lost!

Do you wish to continue? y/n: [n]: y

.
Receiving c2900-universalk9-mz.SPA.151-4.M4.bin from 10.10.10.10
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!

! truncated
program flash location 0x61fe0000
program flash location 0x61ff0000
program flash location 0x62000000
```

You've recovered the system image, the only thing left to do after the download has completed is:

```
rommon 8 > reset
```

IOS Image Upgrade

22) Verify SW1 is running C2960 Software (C2960-LANBASE-M), Version 12.2(25)FX

```
SW1#sh version
Cisco IOS Software, C2960 Software (C2960-LANBASE-M),
Version 12.2(25)FX
```

23) Use the TFTP server to upgrade to c2960-lanbasek9-mz.150-2.SE4.bin

```
SW1#copy tftp flash
Address or name of remote host []? 10.10.10.10
Source filename []? c2960-lanbasek9-mz.150-2.SE4.bin
Destination filename [c2960-lanbasek9-mz.150-2.SE4.bin]?

Accessing tftp://10.10.10.10/c2960-lanbasek9-mz.150-
2.SE4.bin....
Loading c2960-lanbasek9-mz.150-2.SE4.bin from 10.10.10.10:
!!!
```

24) Reboot and verify the switch is running the new software version.

```
SW1#reload
Proceed with reload? [confirm]

SW1#show version
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M),
Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
```